Eric Asare

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Education

New York University (NYU), USA Aug 2026 – Present
Doctor of Philosophy in **Computer Science**

New York University Abu Dhabi (NYUAD), UAE Sep 2020 – Dec 2024
Bachelor of Science in **Computer Science**, Minor: Interactive Media
GPA: 3.879/4.0

Honors: NYU Founders' Day Award - University Honors Scholar distinction for top-ranking baccalaureate graduates.

Coursework: Honors Programming Languages, Design and Analysis of Algorithms II, Cloud and Machine Learning

Research Experience

Software Analysis and Developer Support (SANAD) Lab, NYUAD Feb 2025 – Aug 2026
Junior Research Scientist (*PI: Dr. Sarah Nadi*)

- Conducted a large-scale empirical study of how third-party Python libraries adopt and evolve type annotations, analyzing 11,021 PyPI libraries. Paper accepted at *MSR 2026*.
- Built on this work with an empirical study of how library type annotations affect downstream Python clients, analyzing 13,751 client repositories and 4,178 libraries. Paper accepted at *ASE 2026*.
- Contributed to a controlled study evaluating ChatGPT's impact on students' computational thinking. A short paper was accepted at *EASE 2025*, followed by a full paper at *CHASE 2026*.

Center for Technology and Economic Development (CTED), NYUAD May 2024 – Aug 2024
Research Assistant (*PI: Dr. Yaw Nyarko*)

- Investigated payment methods offered by third-party apps used by Ghanaian farmers and visualized regional app usage data in MATLAB.
- Tested blockchain-based research software for farmers and implemented app alerts to improve usability.
- Supported finance, fundraising, and website work for school, water, and heritage projects in Kumawu.

Tile Size Selection Using a Hill-Climbing Algorithm, NYUAD Sep 2023 – May 2024
Final Year Capstone (*PI: Dr. Riyadh Baghdadi*)

- Developed a hill-climbing algorithm to select tile sizes for the MLIR compiler.
- Evaluated tile-size selections on benchmarks to identify effective configurations.
- Presented the findings in a report, poster, and oral presentation.

Publications

Eric Asare, Luca Di Grazia, and Sarah Nadi (2026). "All or Nothing: How Library Type Annotations Affect Client-Side Errors in Python". In: *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering*. ASE '26. Association for Computing Machinery, pp. 1308–1319. ISBN: 9798400728822. DOI: 10.1145/3832783.3834434. URL: https://doi.org/10.1145/3832783.3834434.

Eric Asare and Sarah Nadi (2026). "How do third-party Python libraries use type annotations?" In: *Proceedings of the 23rd International Conference on Mining Software Repositories*. Association for Computing Machinery, pp. 37–48. ISBN: 9798400724749. URL: https://doi.org/10.1145/3793302.3793337.

May Mahmoud, **Eric Asare**, Nisa Shahid, Nourhan Sakr, and Sarah Nadi (2026). "Fast and Fleeting: Evaluating ChatGPT's Impact on Students' Computational Thinking Skills". In: *Proceedings of the 19th International Conference on Cooperative and Human Aspects of Software Engineering*. CHASE'26. Association for Computing Machinery, pp. 64–74. ISBN: 9798400724848. DOI: 10.1145/3794860.3794894. URL: https://doi.org/10.1145/3794860.3794894.

May Mahmoud, **Eric Asare**, Nourhan Sakr, and Sarah Nadi (2025). "A friend or a foe? Evaluating ChatGPT's Impact on Students' Computational Thinking Skills". In: *Proceedings of the 2025 29th International Conference on Evaluation and Assessment in Software Engineering Companion*. EASE Companion '25. Association for Computing Machinery, pp. 176–181. ISBN: 9798400718328. DOI: 10.1145/3727967.3756830. URL: https://doi.org/10.1145/3727967.3756830.